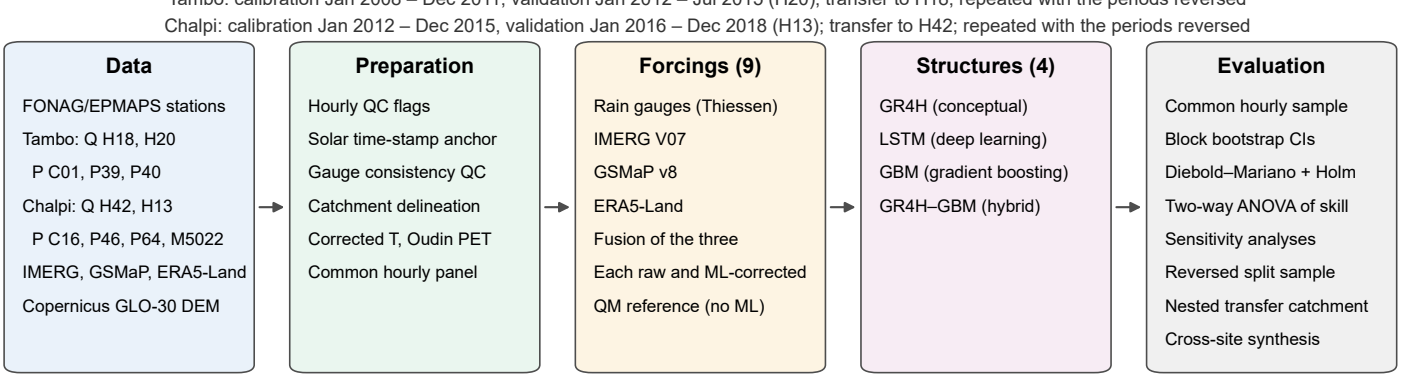
Tambo: calibration Jan 2008 – Dec 2011, validation Jan 2012 – Jul 2015 (H20); transfer to H18; repeated with the periods reversed
Chalpi: calibration Jan 2012 – Dec 2015, validation Jan 2016 – Dec 2018 (H13); transfer to H42; repeated with the periods reversed
Data
FONAG/EPMAPS stations
Tambo: Q H18, H20
P C01, P39, P40
Chalpi: Q H42, H13
P C16, P46, P64, M5022
IMERG, GSMaP, ERA5-Land
Copernicus GLO-30 DEM
Preparation
Hourly QC flags
Solar time-stamp anchor
Gauge consistency QC
Catchment delineation
Corrected T, Oudin PET
Common hourly panel
Forcings (9)
Rain gauges (Thiessen)
IMERG V07
GSMaP v8
ERA5-Land
Fusion of the three
Each raw and ML-corrected
QM reference (no ML)
Structures (4)
GR4H (conceptual)
LSTM (deep learning)
GBM (gradient boosting)
GR4H–GBM (hybrid)
Evaluation
Common hourly sample
Block bootstrap CIs
Diebold–Mariano + Holm
Two-way ANOVA of skill
Sensitivity analyses
Reversed split sample
Nested transfer catchment
Cross-site synthesis